Positive source coordinates for a smaller Diophantine consistency candidate

Nine positive variables, exact degree 1 797 796,
465 arithmetic operations, and a compact coefficient tuple

Research contribution prepared for Vladimir Reshetnikov's ProveIt repository

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Abstract

We give an explicit refinement of the latest nine-positive-variable construction in the reviewed ProveIt snapshot. Two positive coordinate changes in its quadratic source reduce the weighted source from 89 to 86 records. Repacking that source lowers the largest coordinate exponent from 116 to 100 and the convolution height from 2161 to 2047. Composing with the existing numerical ratio and norm reduction yields one fixed integer polynomial of exact total degree 1 797 796, evaluated by a 465-operation division-free circuit. The comparison construction has degree 1 910 404 and 473 operations. The new equation also admits the complete compact tuple already established in package 44, whose four native payloads total 295,388 raw bytes.

The archive contains literal polynomial data, deterministic builders, materialized native coefficients, exact algebra and isolation checks, degree certificates, inherited proofs and licenses, and a secondary 15-positive-variable degree-24/89-operation refinement. These are incremental advances. We do not establish a global simplest equation, a new variable bound, or a construction independent of K215 and Jones. The new local equivalences and arithmetic construction are proved here; the semantic calibration remains inherited. In particular, this contribution does not close the repository's explicitly outstanding object-language ZFC internalization obligation.

1 Result, comparison, and scope

The primary equation is

$$\boxed{F_9 = \left[\Theta^2 - \Omega^2(a_0 + 64a_1)\right]^2 - 256\Omega^4 a_0 a_1 = 0, \qquad (b, g, h, s, w, C, i, j, n) \in \mathbb{Z}^9_{>0}.} \tag{1}$$

Every symbol on the right is a polynomial abbreviation defined below. The only unknowns are the nine displayed inputs. All exponents are fixed nonnegative integers. No variable exponent, divisibility condition, bitwise operation, root, or quantifier is hidden in the equation. Such operations occur only in its proof.

The primary machine-readable specification is `research_nine/compact_candidate/generated/gap_100_positive.json`. Every nonliteral fixed leaf has a polynomial recipe in four specified integers, or is a specified power of two. The native integers are supplied as authenticated unsigned little-endian payloads and by a deterministic reconstruction program. Thus no universal-polynomial parameter remains unspecified.

The baseline is package 59's `nine_free_adjunct` construction at the repository snapshot `beb0192cb8dc36f78c369c080f2adfbc1887a486`. We use its cost convention: one operation for every binary addition, subtraction or multiplication, including scalar multiplications and the multiplications implementing fixed exponents of polynomial registers. Reuse and access to fixed integer coefficients are free. This is arithmetic circuit size, not bit complexity.

Construction	Positive variables	Exact degree	Operations	Port bytes
Package 59, free carrier	9	1,910,404	473	533,099
New source, original tuple	9	1,797,796	465	533,099
New source, compact tuple	9	1,797,796	465	295,388
Package 58, integral-root form	15	24	90	295,388
Supplement: shifted transport	15	24	89	295,388

Table 1: Measured improvements. Raw bytes count the four native ports, not the expanded polynomial's coefficients. The compact native tuple is inherited from package 44. Its use here is a new coherent composition, not a new native coefficient-size record.

The degree decreases by 112 608, approximately 5.89446%; the circuit saves eight operations, approximately 1.69133%. Using the compact tuple reduces native raw bytes by approximately 44.5904%. Both complete specializations are retained separately. The natural-domain translation

$$F_9^{\mathbb{N}}(x_1, \ldots, x_9) = F_9(x_1 + 1, \ldots, x_9 + 1), \qquad \mathbb{N} = \{0, 1, \ldots\},$$

has the same exact degree and a 472-operation supplied circuit. No nine-unrestricted-integer claim is made.

1.1 The proof boundary

There are three different levels.

1. **New algebra and encoding.** The positive source substitutions, the 86-record identities, finite code, range and valuation calibration, and exact circuit metrics are established here. With the explicitly stated numerical Jones ratio theorem, they identify positive zeros of F_9 with common positive zeros of the specified quadratic source.
2. **Inherited semantics.** The native compiler, Pell arguments, Wang943 language and calibrated K215 endpoint identify the intended source with a finite syntactic ZFC refutation. Those are inherited mathematical dependencies. We checked the new adapter and literal native data, not the entire transitive semantic proof chain.
3. **Internal ZFC derivation.** The reviewed repository's `OBJECT_ZFC.md` explicitly leaves the actual internal equivalence derivation open. An external Lean theorem about the `BProv` predicate is not itself a derivation of that theorem inside the object calculus. We supply neither a new object-language derivation nor a new end-to-end Lean/Rocq certificate.

The fully established new result is therefore a smaller explicit arithmetic candidate preserving the inherited mathematical semantics. If the inherited source calibration and the imported arithmetic/compiler correctness statements are established internally in ZFC, the reductions transport them to the intended statement

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^9 \; F_9(\mathbf{x}) = 0 \iff \mathrm{Con}(\mathrm{ZFC})\right]. \tag{2}$$

This report does not present (2) as a newly exported internal theorem. A missing machine certificate alone would not refute an ordinary mathematical in-ZFC proof; an external standard-natural equivalence alone does not supply one either. No consistency assumption is used, and neither side of the equivalence is asserted to be provable in ZFC.

2 The fixed compact coefficient instance

Use the entire native tuple from package 44:

$$\beta = B_* - 1, \qquad K_0 = D_C + B_* D_R, \qquad M_C, \qquad M_F = M_F^{\text{native}} + B_* - 1, \qquad B_* = 2^{590069}.$$

The native radix B_* is a fixed integer, distinct from the packing radix $B = a_* b^2$ introduced later. The full native proof and data are in `upstream44/native/`.

Port	Bit length	Raw bytes	Gzip bytes
β	590,069	73,759	108
K_0	592,884	74,111	8,940
M_C	590,069	73,759	193
M_F, already shifted	590,070	73,759	204
Total		295,388	9,445

The binding aliases `K` to K_0, and `MF_source` to M_F. No individual port may be substituted from another tuple.

For an explicit finite recipe, put $\rho = 2^{67}$, $\mu = 2^{51} - 2$. Enumerate the literal ordered 943-tile list by $i = 0, \ldots, 942$. For tile $i = (w_i, n_i, e_i, s_i)$, set

$$v_i = w_i + 2^8 n_i + 2^{17} e_i + 2^{25} s_i + 127 \cdot 2^{34}, \qquad c_i = 1 + 2^{10}(2^{41} - v_i).$$

Adjoin copy positions $1023 + j, 1013 + j, 1003 + j, 983 + j$, with respective bit-position lists

$$(0, \ldots, 7, 34), \quad (8, \ldots, 16, 35),$$
$$(17, \ldots, 24, 36), \quad (25, \ldots, 33, 37).$$

If a copy position has bit position p, set its $c_e = 2^{10+p}$. Adjoin $c_{1095} = 2^{48}$, $c_{2195} = 2^{49}$, $c_{3296} = 2^{50}$, and $c_{1063} = c_{1064} = 0$. Let $\mathcal{E}$ be the resulting 986-position support, including the two zero-valued positions. Then

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$
$$D_R = \rho^{42},$$
$$Z_* = (2^{65} - 1) \sum_{e=8792}^{8804} \rho^e + (2^{19} - 1)\rho^{8805},$$
$$M_C = B_* - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \neq 1}} \rho^e - 2\rho^{1063},$$

$$M_F^{\text{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{8} \rho^{1065+j} + \rho^{2195} + \rho^{3296} + Z_*.$$

The tile list is fixed archive data, not an existentially chosen program. The native builder reconstructs it from the 430 literal K215 transitions.

Replaying the independent native auditor checks 12 pinned source files, 66,062 native basis interactions, the literal transitions, 38,663 one-bit mutations, the rank controller and exact payloads. Its 87,380 small complement models supplement the unbounded proof. The rank-control prime is 581197, with dimensions $67 \cdot 8807 = 590069$. These finite checks do not replace the native semantic theorem.

3 Normalize positive source coordinates before packing

The source has 31 positive coordinates and 22 quadratic residuals. Names in this section refer to the source, independently of similar names in the final numerical tail. The new coordinate order is

$$\begin{aligned}(&J, F, \alpha, z, f, h, i, o, j, s, w, t, \eta, \zeta, y, \gamma,\\ &N, E, p, c, a, X, D, C_2, S, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).\end{aligned} \tag{3}$$

Here t, p, S replace τ, P, L, respectively.

3.1 Two Pell coordinates become positive gaps

Let $k = \eta + \zeta$. The old source includes

$$a - E = sN, \quad c - \eta = k(a - E), \quad P - k = E(c - \eta), \quad \tau^2 = P(P - k) + 1.$$

The first two rows force $a - E > 0$, $c - \eta > 0$. Consequently

$$p := P - k = E(c - \eta) > 0, \qquad \tau^2 = p^2 + kp + 1 > p^2, \qquad t := \tau - p > 0.$$

The changed rows are

$$\boxed{p - Ec + \eta E = 0, \qquad t^2 + 2pt - \eta p - \zeta p - 1 = 0.} \tag{4}$$

Conversely, positive p, t give positive $P = p + \eta + \zeta$, $\tau = p + t$, satisfying precisely the old rows. Neither old variable occurs elsewhere. This is a bijection of the complete positive common zero sets, not a restriction to selected canonical witnesses.

The first row loses two weighted monomial records. The second retains five, but replaces the active square P^2 by the pair pt. That change in the active pair pattern enables a better packing. Coordinate normalization before packing is the main new ingredient.

3.2 Supply the square instead of the square minus one

Write $q = \beta J + 1$. Replace the old register $L = q^2 - 1$ by $S = L + 1$. The first two rows become

$$S - \beta^2 J^2 - 2\beta J - 1 = 0, \qquad N - S - \beta J S = 0. \tag{5}$$

The inverse is positive: the first row gives $S = q^2 \geq 4$, so $L = S - 1 > 0$.

Retain the inherited invertible row operations

$$\rho_{14} = r_{14} + r_2, \qquad \rho_{15} = r_{15} + r_3, \qquad \rho_j = r_j \ (j \neq 14, 15).$$

Under $L = S - 1$, row 1 gains one record, row 2 loses two, and row 14 keeps its count. Its $-FL$ contributes $+F$, cancelling the old $-F$, while $-\alpha L$ contributes $+\alpha$. No quadratic record changes in this second step. Thus

$$89 \longrightarrow 87 \longrightarrow 86 = 7 \text{ constant} + 36 \text{ linear} + 43 \text{ quadratic records},$$

representing 39 distinct quadratic monomials.

3.3 The transport convention and coherent rebinding

The current package-31 source and native package 44 use the shifted positive transport coordinate z_+:

$$(K_0 + w)C = F + z_+(q-1), \qquad C = q - F - \alpha.$$

Our weighted source uses $z = z_+ + 1$:

$$(K_0 + w)C + (q - F) - z(q-1) = 1.$$

The positive inverse uses inherited source consequences $C > 0$, $F < q$, and $w > q$, giving

$$(z-1)(q-1) = (K_0 + w)C - F > 0, \qquad z \geq 2.$$

This convention is explicit in the independent checker: it substitutes $z_+ = z - 1$, accounting for $+\beta J$ in row 15. No historical byte identity between these two source presentations is claimed.

Package 44, Section 7, proves the same shifted 16-variable Pell system for the compact tuple. Projection of the 31-variable source forgets defining registers. Its canonical Pell witnesses give the reverse positive lift. In particular the delicate register $g = y^2 - V^2$ obeys

$$(I^2 - 1)g = V^2 - 1,$$

and the canonical witnesses have $I, V > 1$, so $g > 0$. The complete native theorem supplies this lift; it is not assumed for arbitrary assignments without the canonical completeness argument.

4 Literal degree-separated packing

The literal 86 weighted records are in `research_nine/compact_candidate/data/parent_source.json`. All 22 residual formulas, their targets and coordinate weights are also displayed in Appendix A. Their independent reconstruction treats all four ports as independent symbols. In the coordinate order (3), the weights are

$$\begin{aligned}(w_1, \ldots, w_{31}) = (&6, 10, 37, 46, 4, 83, 60, 50, 65, 45, 20, 87, 96, 80, 0, 41,\\ &3, 89, 95, 100, 97, 84, 1, 67, 21, 59, 18, 19, 61, 14, 64).\end{aligned} \tag{6}$$

The zero weight is intentional. Assign target exponents by the paired lists

$$\begin{aligned}\text{rows: } &(2, 18, 9, 20, 19, 7, 12, 17, 11, 10, 1, 15, 13, 21, 4, 16, 8, 3, 22, 5, 14, 6),\\ \text{targets: } &(27, 72, 270, 320, 448, 561, 616, 686, 887, 893, 906, 966,\\ &\quad 1139, 1190, 1255, 1279, 1472, 1499, 1545, 1733, 1822, 2016).\end{aligned}$$

Let q_j be the target of row j, and $W = 100$. The targets are unrelated to the source abbreviation $q = \beta J + 1$.

If $\rho_j(\mathbf{y}) = \sum_\alpha a_{j,\alpha}\mathbf{y}^\alpha$, set

$$c_{j,\alpha} = \begin{cases} 2a_{j,\alpha}, & |\alpha| = 0, 1,\\ a_{j,\alpha}\prod_i \alpha_i!, & |\alpha| = 2,\end{cases}$$

and define the three kernels

$$D_d(T) = \sum_{j,\ |\alpha|=d} c_{j,\alpha} T^{q_j - \sum_i \alpha_i w_i}, \qquad d = 0, 1, 2. \tag{7}$$

All exponents are nonnegative. Expanded lists are supplied in `data/split_kernels.json`.

Lemma 1 (Exact target extraction)**.** *For* $U_{\mathbf{y}}(T) = \sum_i y_i T^{w_i}$,

$$[T^{q_j}](D_2(T)U_{\mathbf{y}}(T)^2 + D_1(T)U_{\mathbf{y}}(T) + D_0(T)) = 2\rho_j(\mathbf{y}).$$

The kernel degrees are $(1545, 1920, 1847)$*, the leading coefficient of* D_2 *is* 1*, and* $K_{\text{code}} = \max_d(\deg D_d + dW) = 2047$*.*

Proof. For each source record and every monomial of its degree, the checker tests whether the shifted exponent reaches any target. The degree $0, 1, 2$ bases have $1, 31, 496$ monomials. All $7 + 36 \cdot 31 + 43 \cdot 496 = 22451$ interactions are enumerated. A hit occurs only for the record's own row and monomial. The factorial normalization compensates for the two copies of cross terms in a square. Exact coefficient-polynomial comparison proves the identities, and the literal nonzero kernel heads give the degrees. □

4.1 Packing registers and bounds

Every weighted coefficient satisfies $|c_{j,\alpha}| < M$, with

$$M = 2^{1182955}, \quad a_* = 4096M = 2^{1182967}, \quad c_* = 2^{9463725}.$$

The exponents come from every exact coefficient recipe and the four port bit bounds. For positive outer b, g, define

$$\begin{aligned}
B &= a_* b^2, & \ell &= \sum_i B^{w_i}, & U_{\text{src}} &= \ell + g, \\
G_{\text{src}} &= (D_2(B)U_{\text{src}} + D_1(B))U_{\text{src}} + D_0(B), & T_{\text{mark}} &= \sum_j B^{q_j}, & N_0 &= B^{101}, \\
\text{Mask} &= N_0 - 1 - (b-1)\ell, & J_{\text{code}} &= 2G_{\text{src}} + T_{\text{mark}}, & & \\
S_{\text{code}} &= g + N_0 J_{\text{code}}, & T_{\text{code}} &= \text{Mask} + N_0(B-2)T_{\text{mark}}. & & (8)
\end{aligned}$$

Finally set

$$\begin{aligned}
\mathcal{L} &= bB^{2149}, & R &= \mathcal{L}(S_{\text{code}} + T_{\text{code}} + 1) - S_{\text{code}} - 1, \\
N &= c_* bB^{1128}, & t_{\text{cap}} &= bB^{100}, & G_{\text{cap}} &= 32t_{\text{cap}}.
\end{aligned} \tag{9}$$

Lemma 2 (Unconditional positivity and the full soundness range)**.** *For every positive* b, g*,* $G_{\text{src}} > 0$*,* $S_{\text{code}}, T_{\text{code}} \geq 0$*,* $R, N \geq 8$*, and* $b \leq R, N$*. If* $g < 4bB^{100}$*, then*

$$0 < g < N_0, \qquad S_{\text{code}} + T_{\text{code}} < B^{2149} \leq \mathcal{L}.$$

Proof. Distinct kernel exponents give

$$D_2(B) > \tfrac{1}{2}B^{1847}, \quad |D_1(B)| < 2MB^{1920}, \quad |D_0(B)| < 2MB^{1545}.$$

Since $U_{\text{src}} \geq B^{100}$, the negative contributions relative to $B^{1847}U_{\text{src}}^2$ are below $2M/B^{27}$ and $2M/B^{502}$, each below $1/8$. Hence $G_{\text{src}} > \frac{1}{4}B^{1847}U_{\text{src}}^2 > 0$. The mask has digits $B - b$ at selected weights and $B - 1$ elsewhere, so lies in $[0, N_0)$. Also $R = (\mathcal{L} - 1)S_{\text{code}} + \mathcal{L}T_{\text{code}} + \mathcal{L} - 1 \geq \mathcal{L} - 1$. On the bounded branch, $\ell < 2B^{100}$ and $U_{\text{src}} < 6bB^{100}$. With $Z = B^{2048}$, geometric summation gives

$$|G_{\text{src}}| < \frac{43Mb^2Z}{B-1}, \qquad \frac{2G_{\text{src}}}{Z} < \frac{86Mb^2}{B-1} < \frac{1}{32}.$$

The maximum target is 2016, so $T_{\text{mark}} < Z/64$, $(B-2)T_{\text{mark}} < Z/64$. As $4b < B$, $g < N_0$, and $J_{\text{code}} + (B-2)T_{\text{mark}} + 2 < Z$. It follows that $S_{\text{code}} + T_{\text{code}} < N_0 Z = B^{2149}$. This proves the whole range delivered by soundness, rather than only the narrower completeness range. □

5 From digit disjointness to one binomial valuation

Assume now $b = 2^k$, allowing $k = 0$, and $B = 2^v$, where $v = 1182967 + 2k$. On the bounded branch, the low-mask condition $g \,\&\, \mathrm{Mask} = 0$ is equivalent to

$$g = \sum_i z_i B^{w_i}, \qquad 0 \le z_i < b.$$

Put $y_i = z_i + 1$. Then $1 \le y_i \le b$ and $U_{\mathrm{src}} = \sum_i y_i B^{w_i}$. Every coefficient h_u of the source convolution satisfies

$$|h_u| < M\big((\sum_i y_i)^2 + \sum_i y_i + 1\big) \le 993 M b^2 < B/4.$$

At a target, the signed base-B digit of J_{code} is $2h_u + 1$, with incoming borrow -1 or 0. Its normalized digit avoids the mask $B - 2$ precisely when $h_u = 0$: zero gives digit 0 or 1, positive h_u gives a digit between 2 and $B/2$, and negative h_u gives a digit above $B/2$. Lemma 2 ensures finite, nonoverlapping words. Lemma 1 therefore proves

$$S_{\mathrm{code}} \,\&\, T_{\mathrm{code}} = 0 \quad \Longleftrightarrow \quad g \text{ encodes a positive common source zero.} \tag{10}$$

Conversely, every positive source zero is encodable for every sufficiently large binary b, with $0 < g < bB^{100}$, hence $32g < G_{\mathrm{cap}}$. The strict lower bound follows because the all-one source tuple is not a zero: its fifth residual is 1.

For $\mathcal{L} = 2^u$ and $S_{\mathrm{code}} + T_{\mathrm{code}} < \mathcal{L}$, the two blocks

$$R = (S_{\mathrm{code}} + T_{\mathrm{code}})\mathcal{L} + (\mathcal{L} - 1 - S_{\mathrm{code}})$$

give

$$\mathrm{pop}(R) = u + \mathrm{pop}(T_{\mathrm{code}}) - \tau_{\mathrm{carry}}, \qquad \tau_{\mathrm{carry}} = \mathrm{pop}(S_{\mathrm{code}}) + \mathrm{pop}(T_{\mathrm{code}}) - \mathrm{pop}(S_{\mathrm{code}} + T_{\mathrm{code}}) \ge 0.$$

The defect vanishes exactly when $S_{\mathrm{code}} \,\&\, T_{\mathrm{code}} = 0$. Counting the low mask and the 22 target masks gives

$$\mathrm{pop}(T_{\mathrm{code}}) = 123v - 31k - 22, \qquad u = k + 2149v.$$

Consequently

$$\begin{aligned} u + \mathrm{pop}(T_{\mathrm{code}}) &= 2272v - 30k - 22 \\ &= 2(9463725 + k + 1128v) = 2\log_2 N, \end{aligned}$$

using $2 \cdot 9463725 = 16 \cdot 1182967 - 22$. Legendre's formula $\nu_2\binom{2R}{R} = \mathrm{pop}(R)$ then yields

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_{\mathrm{code}} \,\&\, T_{\mathrm{code}} = 0. \tag{11}$$

The boundary $b = 1$ is included, but cannot produce a final zero: then $\mathrm{Mask} = N_0 - 1$, while $0 < g < N_0$. The calibration makes no unproved restriction to a sparse family of binary bases.

6 The complete nine-variable polynomial

For the nine positive inputs in (1), put $j_0 = j - 1 \geq 0$. Use (8)–(9), and define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R + 1 + h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & J &= 4A - 5,\\
Q &= 3W_1 C - 2(W_1^2 - 1), & D &= (A^2-1)C^2 + 1, & k_0 &= iJC,\\
E &= k_0 DC, & F &= (A^2-1)E^2 + 1, & & \\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + j_0 C), & & \\
I &= (G^2-1)H^2 + 1, & a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI.
\end{aligned} \tag{12}$$

Set

$$\begin{aligned}
m &= K^2(4t_{\mathrm{cap}} - g) - 16t_{\mathrm{cap}}(C - KY)^2, & & & \\
V_0 &= PK, & T_{\mathrm{pair}} &= k_0(8D^2 - 4D - 1), & V_1 &= T_{\mathrm{pair}} GH,\\
\Omega &= Jm, & E_0 &= J(n + W_1^2) + Q, & \Theta &= E_0 - \Omega(V_0 + V_1).
\end{aligned} \tag{13}$$

Then F_9 is exactly (1). The fixed kernels, source records, coefficient tuple and the displayed definitions specify the entire polynomial. No intermediate register adds a witness.

6.1 The inherited numerical ratio interface

We use the numerical Jones ratio soundness and completeness theorem in the exact reparameterized form retained by package 59. Under the unconditional hypotheses in Lemma 2 and the recovered carrier range $C > B_1 + W_1$, the conditions

$$a_0 \text{ square}, \qquad a_1 \text{ square}, \qquad J \mid Q, \qquad m > 0$$

imply binary b, $8g < G_{\mathrm{cap}}$, and $N^2 \mid \binom{2R}{R}$. Conversely, the encoded source witnesses with $32g < G_{\mathrm{cap}}$ admit the required ratio witnesses. This is an imported unbounded arithmetic theorem, not a consequence of the finite audits.

Its CRT coordinate substitution is exact. The previous condition is

$$JF \mid QF + J(H_{\mathrm{old}} - C), \qquad H_{\mathrm{old}} = B_1 + j_{\mathrm{old}} C, \quad j_{\mathrm{old}} \geq 1.$$

Since $F \equiv 1 \pmod{J}$, $\pmod{C}$, it is equivalent to $J \mid Q$ and $F \mid H_{\mathrm{old}} - C$. The positive quotient is $B_1 + j_0 C$ for a unique $j_0 \geq 0$. Conversely,

$$j_{\mathrm{old}} = 1 + \frac{F-1}{C} B_1 + F j_0$$

is a positive integer giving exactly the displayed H. The division occurs only in the proof; $C \mid F - 1$. The positive input must therefore be $j = j_0 + 1$, rather than silently imposing $j_0 > 0$. The normalized margin satisfies

$$8m = [K^2 - 4(C - KY)^2] G_{\mathrm{cap}} - 8gK^2.$$

7 The norm has no extraneous positive zeros

The logical order is crucial: establish the sign of m before assuming a carrier range or integral roots.

Lemma 3 (Unconditional strict majorants)**.** *For every positive input assignment,*

$$V_0^2 - a_0 = K^2 - 1 > 0, \qquad T_{\text{pair}}^2 - 64DF = (8D+1)k_0^2 - 64D > 0,$$

and

$$\begin{aligned}\Delta_1 := V_1^2 - 64a_1 = [(8D+1)k_0^2 - 64D](GH)^2 \\ + 64DF(H^2-1) > 0.\end{aligned} \tag{14}$$

Also $E_0 > 0$.

Proof. The unconditional bounds imply $A \geq 2$, $J \geq 3$, $D \geq A^2$, $D > C^2$, $F \geq A^2$, $G > A$, $H > 1$, and $k_0 \geq 3$. For the second identity substitute $F = 1 + k_0^2 D^2(D-1)$ and expand $(8D^2 - 4D - 1)^2$. Its difference is at least $8D + 9$. The third identity follows from $I = (GH)^2 - H^2 + 1$. Finally $A > W_1$, and

$$E_0 = Jn + (J-2)W_1^2 + 3W_1C + 2 > 0,$$

including assignments where the free carrier makes Q negative. □

Theorem 4 (Arithmetic zero equivalence)**.** *With the coherent fixed coefficients and the stated numerical ratio theorem,* F_9 *has a positive zero exactly when the new 31-positive-coordinate quadratic source has a common zero.*

Proof. Suppose $F_9 = 0$. If $m = 0$, then $\Omega = 0$ and $F_9 = E_0^4 > 0$, a contradiction. Otherwise the real norm factorization supplies signed real roots $r_0^2 = a_0$, $r_1^2 = 64a_1$ with

$$n + W_1^2 + \frac{Q}{J} = m(V_0 + V_1 + r_0 + r_1).$$

The left side is positive. The strict majorants make the right parenthesis positive for every sign choice. Thus $m > 0$, without an integrality assumption.

The margin now gives $g < 4t_{\text{cap}}$ and $|C - KY| < K/2$. As $Y \geq 64$,

$$C > K(Y - \tfrac{1}{2}) > 2K \geq 2(P+R) > 2R + 1 + W_1.$$

Here $K \geq P + R$ and $P > U \geq W_1$. Also $A^2 > P$, $D > C^2$, $F \geq A^2$, $G \geq A$, and $I \geq G^2$, giving

$$a_1 = DFI > C^2A^4 > K^2P^2 > a_0.$$

The rational sum $x = r_0 + r_1 = \Theta/\Omega$ is nonzero. The identity $2xr_0 = x^2 + a_0 - 64a_1$ makes both roots rational, hence integral because their squares are integers. Moreover $r_1/8$ is a rational square root of a_1, hence an integer. The factor equation implies $J \mid Q$. Ratio soundness now supplies binary b, $8g < G_{\text{cap}}$, and binomial divisibility. Equations (11) and (10) recover the positive source zero.

Conversely, encode a positive source zero with sufficiently large binary b, giving $32g < G_{\text{cap}}$ and the binomial condition. Ratio completeness supplies witnesses and integral roots. Transport the CRT coordinate as above and choose positive roots. Since $W_1^2 < D$, $W_1C < D$, $Q/J \leq W_1C$, and $T_{\text{pair}} \geq 9D$, one has

$$W_1^2 + \frac{Q}{J} < 2D < V_1.$$

Thus

$$n = m(V_0 + V_1 + r_0 + r_1) - W_1^2 - \frac{Q}{J}$$

is a strictly positive integer. It makes one norm factor vanish, so $F_9 = 0$ in the stipulated domain. □

The source changes of Section 3 and the coherent native adapter transport this equivalence to the inherited semantics, at the proof level stated in Section 1.

8 Exact degree, rather than naive degree propagation

The main circuit has material leading cancellation. The certificate uses the exact identity

$$\begin{aligned}\Theta^2 - \Omega^2(a_0 + 64a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) \\ + \Omega^2[(K^2 - 1) + \Delta_1 + 2V_0V_1].\end{aligned} \tag{15}$$

Fixed integers have degree zero. On the all-inputs diagonal, the key source degrees and heads are:

Register	Exact degree	Diagonal leading coefficient
U_{src}	200	a_*^{100}
G_{src}	4094	a_*^{2047}
S_{code}	4296	$2a_*^{2148}$
T_{code}	4236	a_*^{2118}
$\mathcal{L}$	4299	a_*^{2149}
R	8595	$2a_*^{4297}$
N	2257	$c_*a_*^{1128}$

The decisive tail degrees are

	A	D	k_0	F	G	H	a_0
deg	17625	35252	17627	141010	176263	149605	122942

and

	a_1	m	Ω	V_0	V_1	Δ_1
deg	827998	70703	88328	61471	413999	722242.

The term $\Omega^2\Delta_1$ uniquely dominates (15), at degree 898898. Its square has degree 1797796, whereas the final subtracted term $256\Omega^4 a_0 a_1$ has degree only 1304252. The exact final diagonal leading coefficient is

$$\boxed{2^{186}a_*^{898488}c_*^{400} > 0.} \tag{16}$$

The certificate supplies a matching multivariate upper bound, and the nonzero diagonal term attains it. The total degree is therefore exactly 1 797 796.

For this calibrated family,

$$\deg F_9 = 104 \deg R + 400 \deg N + 8W + 316 = 816K_{\mathrm{code}} + 1224W + 5044.$$

The reduction decomposes as

$$816(2161 - 2047) + 1224(116 - 100) = 112608.$$

This gives a useful search objective: coordinate design and packing height should be optimized together, since reducing the monomial count alone need not reduce the final degree.

The primary positive evaluator has 159 additions, 22 subtractions and 284 multiplications, all live. The cancellation-resolving circuit is a separate certificate, not the counted primary evaluator. The natural translation has 167 additions, 21 subtractions and 284 multiplications. Its internal j_0 is just the natural j-input, accounting for the subtraction saved.

9 A small degree-24 companion

The companion has 15 positive unknowns, exact degree 24 and 89 operations, compared with package 58's 90. The substantial degree reduction is inherited from package 58; this contribution saves one transport operation. The old shared-register residual is

$$T_{\mathrm{old}} = q\{K_0C + (q - F) - z_{\mathrm{old}}(q - 1)\} + XC - q.$$

Putting $z_{\mathrm{old}} = z + 1$ gives exactly

$$\boxed{T_{\mathrm{new}} = C(qK_0 + X) - q\{F + z(q - 1)\}.} \tag{17}$$

The inherited source inequalities give the positive inverse via $(z_{\mathrm{old}}-1)(q-1) = (K_0+w)C-F > 0$. An eight-operation residual becomes a seven-operation residual. The whole sum of squares has 47 multiplications, 27 additions and 15 subtractions. Its unchanged degree-12 residual ensures exact degree 24 after squaring. The supplement includes both literal circuits, a complete proof and an exact symbolic identity/degree audit. This is not a degree-22 result or a variable-count improvement.

10 Alternative substrates and the limits of the present search

The construction problem does not logically require a particular Turing machine, nor a particular universal-polynomial presentation. We examined alternatives to the inherited substrate as well as algebraic changes to its final arithmetic kernel. None of the independent-substrate routes examined here produced a smaller, fully instantiated polynomial with a complete correctness argument. This is a statement about the work completed in this iteration, not an impossibility theorem or a claim that the existing substrate is optimal.

10.1 A direct proof-checker route

Fix an effective proof calculus for ZFC and an explicit coding of its formulas and finite proofs. The predicate $\mathrm{Ref}_{\mathrm{ZFC}}(p)$, asserting that p codes a refutation, can be chosen primitive recursive. A direct compiler could therefore replace the machine simulation by a polynomial $D(p, \mathbf{x})$ with a verified representation theorem

$$\mathrm{ZFC} \vdash \big(\exists \mathbf{x} \in \mathbb{N}^m \; D(p, \mathbf{x}) = 0\big) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}(p).$$

After existentially quantifying p, the absence of zeros would be equivalent to syntactic Con(ZFC). This is a valid architecture that makes no reference to K215.

To turn that architecture into the requested fixed equation, the following data and arguments must actually be supplied. The syntax encoding must specify formula parsing, bound-variable handling and substitution; axiom recognition must include the Separation and Replacement schemes; and the proof checker must specify every allowed inference. The arithmetic compiler must then emit polynomial syntax, literal fixed coefficients and a proof relating its integer witnesses to the checker. Finally, the checker must be calibrated against the same effective ZFC

calculus used in the consistency assertion. It is the combined cost of these components that should be compared with the present construction.

The MRDP theorem establishes the availability of a Diophantine representation, but an unspecified representation or an unspecified universal-polynomial parameter is not an explicit coefficient instance. Likewise, a theorem accepting supplied representation syntax does not by itself construct that syntax for a particular proof checker. The current ProveIt `QUARTIC_COMPILER.md` explicitly distinguishes these two interfaces. Closely related distinctions between mathematical construction, polynomial data and formal verification also occur in the Isabelle universal-pair development.[1]

A finite logical presentation or a smaller equiconsistent theory could be a useful source language. However, counting axioms does not measure the size of the compiled polynomial. Such a proposal must provide effective proof translations sufficient to justify the intended consistency equivalence, and must still pay for its syntax checker. Replacing proofs by finite models of ZFC would not work: ZFC itself proves that its Infinity axiom rules out finite models. Replacing the theory by an assertion that a model of ZFC exists also requires a separate strength analysis; that assertion cannot simply be treated as an inexpensive equivalent axiom.

10.2 Generic degree reduction is not a new small quartic

An arithmetic circuit can be converted to quadratic equations by introducing values for intermediate registers, and a sum of their squares then has degree at most four. This exchanges degree for unknowns. Our straightforward lift of the ordinary Pell kernel did not improve the existing source-31 point: that source already has 31 positive unknowns and 22 quadratic residuals. Its comparison evaluator has 60 operations and its quartic sum of squares has 125 operations. These figures are the existing `31-sun10/quartic31/QUARTIC60_PROOF.md` benchmark, not a new result of this report.

One illustrative failed elimination concerns $X = wq$. Combining it with the transport relation can remove the letter w from a polynomial identity, but generally leaves only $q \mid XC$. Restoring the old witness $w = X/q$ would require $q \mid X$. It is invalid to cancel C without first establishing $\gcd(q, C) = 1$. In the contemplated route, the desired coprimality depended on source-decoding consequences that themselves used $q \mid X$. We therefore did not accept this as a reduction in the number of variables. This identifies a missing argument in that particular route; it does not rule out a different elimination.

10.3 Two auxiliary congruence routes that do not lower degree

The degree-24 integral-root construction has an auxiliary Pell norm

$$UV^2 - (U-1)y^2 = 1, \qquad U = I^2 = \Delta(f^2-1), \qquad y = V + u.$$

Here $\deg U = 6$, $\deg V = 5$, and the supplied gap u has degree one. The gap residual

$$(V+u)^2 - Uu(2V+u) - 1$$

has degree 12, giving degree 24 after squaring. We examined whether canonical congruences permit a cheaper replacement of the gap.

[1] Jonas Bayer and Marco David, *A Formal Proof of Complexity Bounds on Diophantine Equations*, https://arxiv.org/abs/2505.16963; the associated development is https://isa-afp.org/entries/Diophantine_Universal_Pairs.html.

For the canonical odd Pell index $R \equiv 3 \pmod 4$, $y = \psi_I(R) \equiv -1 \pmod{I^2}$. Thus $t = (y+1)/U$ is a positive integer. The exact identity

$$\begin{aligned} UV^2 - (U-1)(Ut-1)^2 - 1 \\ = U\big[V^2 - U(U-1)t^2 + 2(U-1)t - 1\big] \end{aligned}$$

gives a valid alternative residual when completeness uses these canonical witnesses. Its degree is 14, because of $U(U-1)t^2$, so its square has degree 28. This useful divisibility does not lower the degree of the current gap construction. Supplying Uu instead leads to denominator clearing and likewise did not improve the measured degree.

There is a second canonical congruence. Write $A = a + 2$, so $\Delta = A^2 - 1$, and let $D = \chi_A(R)$, $c = \psi_A(R)$. The odd-subsequence Pell identities, together with $I^2 \equiv 1 - A^2 \pmod{f^2}$, give

$$Ay \equiv -D \pmod{f^2} \qquad (R \equiv 3 \pmod 4).$$

For example, at $R = 3$ one has $\psi_I(3) = 4I^2 - 1$, and multiplication of its reduction modulo f^2 by A gives $-\chi_A(3)$. The general identity follows from the same odd-subsequence recurrence. Notice that reducing the norm alone gives only $(Ay)^2 \equiv D^2 \pmod{f^2}$; the negative branch cannot be selected for arbitrary supplied solutions from that square congruence alone.

The canonical congruence suggests $y = (f^2t - D)/A$. Write $V = f^2W - c$ and center the new quotient by $t = AW + v$. Formally retaining D as a register, after clearing the denominator A^2 and using $D^2 - \Delta c^2 = 1$, the norm numerator is divisible by f^2. The resulting expression still contains $-2A\Delta Wvf^4$. This term has degree 14 under $\deg A = 2$, $\deg \Delta = 4$, $\deg W = 3$ and $\deg v = \deg f = 1$. Moreover, source 58 reconstructs D rather than supplying it as a separate unknown, so that reconstruction would also have to be accounted for in a literal polynomial. This calculation does not provide the hoped-for degree-22 equation. It proves no broader obstruction to using this congruence in a different construction.

10.4 Scope of the reported advance

The demonstrated improvements in this report concern the explicit constructions and comparisons proved in its preceding sections. The alternative proof substrates above remain research proposals. They are not counted as delivered equations, and the calculations that failed to improve degree are retained to make the search boundary clear. In particular, we make neither a global simplicity-minimum claim nor a claim that a construction independent of K215 or Jones is impossible.

Throughout, a proof of ZFC $\vdash$ [no zeros $\leftrightarrow$ Con(ZFC)] is an internal equivalence theorem. It is not a proof in ZFC of either side. Finite coefficient, identity and degree audits establish their stated finite properties; they do not establish the consistency assertion by failing to find a zero.

11 Reproduction, provenance, and integration

From the archive root run

```
python3 checks/verify_all.py
```

to rebuild native coefficients and the nine-variable specializations and replay the independent source, circuit and supplement audits. Mathematical code uses only the Python standard library. No layout search is needed: the supplied finite certificates are the reproduction inputs. Exact polynomial checks are distinguished from modular diagnostic evaluations.

Path	Purpose
`research_nine/compact_candidate/`	Recommended equation, builder, circuit audits and degree certificates.
`research_nine/candidate/`	Separate original-tuple specialization.
`research_alternatives/check_positive_gap.py`	Independent reconstruction from the authenticated source DAG and all target interactions.
`research_low_degree/audit_gap_source.py`	Exact $89 \to 87 \to 86$ source identity audit.
`research_low_degree/supplement15/`	Degree-24/89-operation companion and comparator.
`upstream44/native/`	Complete compact tuple, native proof, pinned sources and licenses.
`upstream31/`	Source-31 data, inherited proofs and original coefficient tuple.
`upstream-status/`	Pinned proof-status documents and external K215 endpoint statement.
`MANIFEST.sha256`	Hashes of distributed files, excluding the manifest itself.

Imported proofs retain historical wording and relative links. Their former frontier claims or coefficient metadata do not override this report's compact binding. In particular, the retained original-tuple parameters $a_* = 2^{2134939}$, $c_* = 2^{17079501}$ appear in the general source proof and comparison. The recommended compact parameters are $2^{1182967}$, $2^{9463725}$. The archive does not replace any certified Lean endpoint or claim a repository Lean build.

11.1 What would constitute a stronger next result

Three concrete directions remain. Further positive coordinate redesign could reduce the active pair constraints, or remove a witness with a proved positive inverse. A direct compiler for sealed ZFC proof codes could address the outstanding internalization problem without the native machine layer, but must emit literal coefficients and an internal correctness argument. A different ratio or finalization mechanism could reduce the large numerical coefficients in $816K_{\mathrm{code}} + 1224W + 5044$, which the present packing improvements leave unchanged.

None of these is asserted as completed. Failing to find a zero, checking finite coefficients, or proving a standard-natural equivalence does not establish the internal consistency statement. The remaining formal target must account for the internal natural numbers of arbitrary ZFC models, including nonstandard witnesses.

A Complete quadratic source

In the following table, $\mathcal{B} = B_{\mathrm{aux}}$ in the coordinate list (3). The generated table is checked against every formal coefficient in the literal source. Its indices $0, \ldots, 30$ correspond in order to the one-based weight list $w_1, \ldots, w_{31}$ in the main text.

A.1 The exact positive quadratic source

All 31 variables in this table range over the positive integers. The four symbols β, K_0, M_C, M_F are the coherent compact coefficient tuple; M_F denotes the port named `MF_source` in its native binding. Write

$$q = \beta J + 1, \qquad k = \eta + \zeta.$$

The source system is $\rho_r = 0$ for $1 \le r \le 22$. The last column gives the target exponent λ_r assigned to that residual in the frozen packing layout. These are local source symbols: for

example, a, c, D, N in this table do not denote the outer encoding constants or the Jones-tail registers.

r	ρ_r	λ_r
1	$S - q^2$	906
2	$N - Sq$	27
3	$X - wq$	1499
4	$a - E - sN$	1255
5	$E - (a - E)X$	1733
6	$c - \eta - k(a - E)$	2016
7	$p - E(c - \eta)$	561
8	$t^2 + 2pt - kp - 1$	1472
9	$D - X - ca - (4a + 3)\gamma$	270
10	$\Delta - (a + 1)(a + 3)$	893
11	$C_2 - c^2$	887
12	$D^2 - C_2\Delta - 1$	616
13	$k + V - hE - jc - 1$	1139
14	$jc - V + \alpha(1 - S) + F(N - S - q + 1)$ $+ (q - 1)(q - N) - J(M_C + M_F q)$	1822
15	$K_0(q - F - \alpha) + X - F - w(F + \alpha)$ $+ (q - 1)(1 - z)$	966
16	$F_2 - f^2$	1279
17	$c + V - fo$	686
18	$g + V^2 - y^2$	72
19	$I - iC_2$	448
20	$\mathcal{B} - I^2$	320
21	$\mathcal{B} + (1 - F_2)\Delta$	1190
22	$y^2 - g\mathcal{B} - 1$	1545

The variable z in row 15 is the original transport variable. The shifted positive transport coordinate used by the native source is $z_{\mathrm{s}} = z - 1$; the positive inverse requires the source-kernel bound $z \geq 2$. The coefficient M_F retains its native binding in this change of variable. The new coordinates are $p = P - (\eta + \zeta)$, $t = \tau - p$, and $S = L_{\mathrm{old}} + 1$. Rows 14 and 15 include the invertible row operations $\rho_{14} = r_{14} + r_2$ and $\rho_{15} = r_{15} + r_3$.

Variable indices and weights. The monomial records use the following zero-based indices and code weights w_i. The unique zero weight is attached to y.

i	variable	w_i	i	variable	w_i	i	variable	w_i
0	J	6	11	t	87	22	D	1
1	F	10	12	η	96	23	C_2	67
2	α	37	13	ζ	80	24	S	21
3	z	46	14	y	0	25	F_2	59
4	f	4	15	γ	41	26	V	18
5	h	83	16	N	3	27	g	19
6	i	60	17	E	89	28	I	61
7	o	50	18	p	95	29	$\mathcal{B}$	14
8	j	65	19	c	100	30	Δ	64
9	s	45	20	a	97			
10	w	20	21	X	84			

Weighted-record convention. For a monomial x^α with $|\alpha| \le 2$, the stored coefficient is $(2-|\alpha|)!\,\alpha!$ times its coefficient in ρ_r. Thus constants, linear terms, and diagonal quadratic terms are doubled; mixed quadratic terms are unchanged. The source has 86 weighted records. The script `write_source_table.py` verifies every displayed residual exactly over $\mathbb{Z}[\beta, K_0, M_C, M_F]$ before writing this table.

B Payload identities and source authentication

The compact raw payload SHA-256 values are:

β:
`0664a868cbfef9869ea05c40a1c445563f685ff857092594c1c1223bc4b21f9d`
K_0:
`c6d523bd32af8dd0345db843cdadf6932bfcad186912a6d112314b60657c554e`
M_C:
`6007cabd5f5228618e8fb613f386d8fc3692a24520d6e4cd72af0262aae1ade4`
M_F:
`48333b0d30a4edeb8579f44286cb59806c5622cb42b367f5b5f9ee827711c737`.

The new 86-record source has SHA-256

`98a3be433694b5c5b5e99020c503cbb2e2a5adf05757b6bbc2357832472700ab`.

The current authenticated 60-operation source has SHA-256

`ac2a67782d344d1e74fd93bc3f09fae0097eaed1829fd8194ae235ae84763e48`,

and Git blob `518b2e3f03c354ce2ecc74f8f7a75e526cd911c7`. The historical package-59 parent hash is not claimed for the new record format. Independent exact reconstruction from the current source removes any reliance on an unavailable historical parent file.

C Reference map

The following sources fix the scope of inherited claims. Archival reference copies and licenses are included; full transitive Lean dependencies remain in ProveIt.

1. ProveIt, reviewed snapshot `beb0192cb8dc36f78c369c080f2adfbc1887a486`, *Computability/DiophantineConsistency*, especially `OBJECT_ZFC.md`, `PROPOSAL_REVIEW.md` and the research collection: https://github.com/VladimirReshetnikov/ProveIt/tree/beb0192cb8dc36f78c369c080f2adfbc1887a486/Computability/DiophantineConsistency.
2. Package 59, *Free-coordinate composition with the degree-separated source*, `nine_free_adjunct/PROOF.md`, and its retained FreeCoordinate proof and review: numerical tail and comparison metrics.
3. Package 31, `quartic31/QUARTIC60_PROOF.md`, `quartic31_system60.json`, and ordinary compiler/Pell proofs: quadratic source and positive transport shift.
4. Package 44, `native/PROOF.md`, `COEFFICIENT_BINDING.json`, `SOURCE_PINS.json`, `build_coefficients.py` and `audit_coefficients.py`: whole compact tuple and native semantic adapter.
5. Package 58, report and `code/build.py`: degree-24 parent.
6. `BusyBeaver.ZFC.KEndpoint215`, `blank_halts_iff_zfc_refutation`: external literal-machine endpoint, distinct from the internal target in (2).
7. J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic 47 (1982), 549–571. The exact numerical ratio interfaces here are the repository's literal versions.
8. J. Bayer and M. David, *A Formal Proof of Complexity Bounds on Diophantine Equations*, ITP 2025, LIPIcs 352, 3:1–3:18; https://arxiv.org/abs/2505.16963v2. Associated AFP development: https://isa-afp.org/entries/Diophantine_Universal_Pairs.html.